

Brock Biology of Microorganisms, 15e (Madigan et al.)
Chapter 25 Microbial Infection and Pathogenesis

25.1 Multiple Choice Questions

- 1) Pathogenicity is the ability
A) of the host to inflict damage on the pathogen.
B) of the host to resist damage by the pathogen.
C) of the pathogen to inflict damage on the host.
D) None of these are correct.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 25.3

- 2) Which of the following is NOT a subunit of lipopolysaccharide (LPS)?
A) O-specific polysaccharide
B) lipid A
C) core polysaccharide
D) M protein

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 25.8

- 3) *Staphylococcus aureus* and *Streptococcus pyogenes* strains can produce _____ that cause the severe and sometimes life-threatening symptoms of toxic shock syndrome.
A) endotoxin
B) superantigen toxins
C) AB toxins
D) cytolytic

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 25.7

- 4) Diphtheria toxin is a
A) type of endotoxin.
B) cytolytic toxin.
C) AB toxin.
D) superantigen toxin.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 25.6

5) Decalcification of the tooth enamel is caused by

- A) lipopolysaccharide.
- B) dextrans.
- C) formic acid.
- D) lactic acid.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.2

6) The following bacterial genera are all implicated in dental caries EXCEPT

- A) *Fusobacterium*.
- B) *Borrelia*.
- C) *Streptococcus*.
- D) *Corynebacterium*.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 25.2

7) Tetanus toxin causes

- A) muscles to be unable to relax.
- B) muscles to be unable to contract.
- C) severe diarrhea.
- D) hemolytic uremic syndrome.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 25.6

8) Shiga toxin is an enterotoxin that is produced by

- A) *Shigella dysenteriae*.
- B) *Clostridium tetani*.
- C) *Escherichia coli* O157:H7.
- D) *Vibrio cholerae*.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.6

9) Exotoxins are _____, but endotoxins are _____.

- A) lipopolysaccharide-lipoprotein complexes / proteins
- B) carbohydrates / proteins
- C) lipids / proteins
- D) proteins / lipopolysaccharide-lipoprotein complexes

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.8

10) An endogenous pyrogen is

- A) a chemical from the bacterial cell wall that causes fever.
- B) a chemical produced by a host's own immune system that causes fever.
- C) a chemical released by a bacterial cell that causes fever.
- D) a chemical produced within a bacterial cell that is released during cell division or at death of the bacterium, at which point it causes fever.

Answer: B

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 25.8

11) *Staphylococcus aureus* produces _____, leading to fibrin clots that protect them from attack by host cells.

- A) collagenase
- B) coagulase
- C) lipase
- D) amylase

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.5

12) Compared with an endotoxin, would a bacterial exotoxin generally be more likely to function after it had been significantly heated?

- A) Yes, because exotoxins are very heat stable compared to endotoxins.
- B) Yes, because exotoxins are proteins and non-living but endotoxins are part of the bacterial cell wall.
- C) No, because exotoxins are carbohydrates and endotoxins are proteins.
- D) No, because exotoxins are heat-labile and less resistant to heating than endotoxins.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.8

13) The α -toxin of *Clostridium perfringens* is

- A) an endotoxin.
- B) an enterotoxin.
- C) a lecithinase.
- D) a superantigen toxin.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.7

14) Which of the following hemolysins is a phospholipase?

- A) lecithinase
- B) streptolysin-O
- C) Staphylococcal α -toxin
- D) leukocidin

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.7

15) *Salmonella* species use _____ to facilitate attachment to gastrointestinal tissue.

- A) type I fimbriae
- B) antiphagocytic proteins
- C) siderophores
- D) pili

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.4

16) Influenza virus targets

- A) respiratory epithelium.
- B) gastrointestinal cells.
- C) oral cavity cells.
- D) throat epithelium.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.1

17) An outer coat consisting of a dense, well-defined polymer layer surrounding a cell and used in attachment is called a

- A) capsule.
- B) cytoplasmic membrane.
- C) lipopolysaccharide layer.
- D) pilus.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.1

18) Capsules are particularly important for

- A) making bacteria more vulnerable to host defense mechanisms.
- B) making bacteria less pathogenic.
- C) protecting bacteria from host defense mechanisms.
- D) allowing bacteria to become more phagocytic.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.3

19) The decrease or loss of virulence of a pathogen is referred to as

- A) aging.
- B) attenuation.
- C) disinfectivity.
- D) lethal dose.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.3

20) It is thought that attenuation occurs especially in a laboratory setting because

- A) nonvirulent or weakly virulent mutants grow faster in laboratory media when there is no selective advantage to virulence.
- B) pathogens lose virulence with age and survive longer in laboratory cultures.
- C) patients can be treated with drugs that induce attenuation.
- D) there is selection for more virulent strains in the laboratory.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.3

21) Which disease CANNOT be prevented via the use of a vaccine generated from an attenuated pathogen?

- A) malaria
- B) measles
- C) mumps
- D) rubella

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.3

22) Which of the following is NOT important for the adherence of bacteria to host tissues?

- A) lipopolysaccharides
- B) capsule
- C) adhesins
- D) slime layer

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.1

23) Which of the following is an important factor in the development of dental caries?

- A) high salt diet
- B) high sugar diet
- C) high acid diet
- D) low salt diet

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.2

24) The spread of pathogens through the blood and lymph systems that results in a bloodborne systemic infection is called

- A) bacteremia.
- B) cancer.
- C) pathogenesis.
- D) septicemia.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.2

25) Which genus produces hyaluronidase?

- A) *Mycobacterium*
- B) *Streptococcus*
- C) *Shigella*
- D) *Procholorococcus*

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.5

26) Which of the following is more likely to cause urinary tract infections?

- A) fimbriated strains of *Escherichia coli*
- B) non-fimbriated strains of *Escherichia coli*
- C) both fimbriated and non-fimbriated strains of *Escherichia coli*
- D) None of the answers are correct.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 25.1

27) Which of the following is NOT an example of an AB toxin?

- A) shiga-like toxin
- B) hemolysins
- C) botulinum toxin
- D) tetanus toxin

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 25.6

28) Which of the following does NOT affect pathogen growth?

- A) availability of microbial nutrients
- B) pH
- C) temperature
- D) All of these answer choices affect growth.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 25.2

29) Siderophores are

- A) molecules that aid directly in attachment to host cells.
- B) inorganic molecules that help in active transport.
- C) organic molecules that help pathogens take up iron.
- D) toxins that affect the gastrointestinal tract.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.4

30) Cytolytic toxins

- A) are extracellular proteins.
- B) cause cell lysis and death.
- C) damage host cytoplasmic membrane.
- D) are extracellular proteins that cause cell lysis and death by damaging the host cytoplasmic membrane.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.7

31) Which of the following is a reason that specific pathogens tend to infect specific tissues?

- A) Some tissues have receptors for microbes, but others do not.
- B) There are chemical and physical differences between tissues.
- C) Tissues with neutral pH are colonized but those with acidic or alkaline pH are not.
- D) Microbes preferentially target external tissues, such as skin, as the environment is more favorable for their growth than inside the body.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 25.2

32) The collective term for the organisms living on or in the human body, as opposed to a general term for organisms in an environmental habitat, is

- A) the human microbiome.
- B) the microbiota.
- C) transient microbial flora.
- D) pathogens.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 24.1

33) _____ in saliva cleaves glycosidic linkages in the peptidoglycan that is present in bacterial cell walls, weakening the wall and potentially causing cell lysis.

- A) Mucus
- B) Lysozyme
- C) Fibrin
- D) Lipid A

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.5

34) Which of the following is a way to prevent attenuation and maintain virulence in a bacterium?

- A) laboratory subculture
- B) inoculating an animal with the bacterium
- C) polymerase chain reaction (PCR)
- D) antibiotic therapy

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 25.4

35) Which of the following is TRUE about flagella?

- A) They are the most important structure in adherence.
- B) They are not involved in adherence and colonization at all.
- C) They are as important in adherence as pili and fimbriae.
- D) They may be involved in adherence to a lesser extent than pili and fimbriae.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 25.1

36) _____ is a damage or injury to a host organism that impairs its function.

- A) Trauma
- B) Infection
- C) Disease
- D) Transmission

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.1

37) Pathogenicity and virulence differ in that

- A) pathogenicity refers to the overall ability of a microbe to cause disease, whereas virulence refers to the ability of one microbe to cause disease relative to another.
- B) virulence refers to the overall ability of a microbe to cause disease, whereas pathogenicity refers to the ability of one microbe to cause disease relative to another.
- C) pathogenicity only refers to pathogens, whereas virulence refers to any microbe that can cause an infection.
- D) virulence only refers to pathogens, whereas pathogenicity refers to any microbe that can cause an infection.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 25.3

38) Epithelial cells lining the respiratory tract form a

- A) dry and rigid barrier with tight junctions between cells.
- B) impenetrable barrier to infection.
- C) mucous membrane.
- D) connective tissue layer.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 23.2

39) What is the difference between adherence and colonization?

- A) Colonization occurs when transient microbes are on a tissue, whereas adherence is a more permanent attachment to cells.
- B) Colonization occurs when microbes begin to spread in host tissues, whereas adherence occurs when microbes initially attach.
- C) Colonization occurs when microbes begin to spread in host tissues, whereas adherence occurs when microbes first begin to reproduce in the host tissues.
- D) Colonization occurs when a microbe begins to grow in host tissues whereas adherence occurs when the microbe initially attaches.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 25.2

40) Which of the following pathogens does NOT use capsules or a slime layer for attachment?

- A) *Vibrio cholera*
- B) *Streptococcus pneumonia*
- C) *Bacillus anthracis*
- D) All of these pathogens use capsules or slime layers for attachment.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 25.1

41) Which of the following is a major growth-limiting micronutrient that influences microbial growth?

- A) iron
- B) water
- C) vitamin K
- D) sugar

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 25.4

42) The dose of an antigen that kills 50% of animals in a test group and is used to estimate the virulence of a pathogen is known as

- A) Antigen dose50.
- B) Virulence-50.
- C) LD50 (lethal dose50).
- D) Death rate-50.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.3

43) The best term to describe the general process by which microorganisms cause diseases is known as

- A) pathogenesis.
- B) virulence.
- C) invasion.
- D) infection.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.1

44) The macromolecules responsible for bacterial adherence that are NOT covalently attached to bacteria are collectively called

- A) lipid A.
- B) biofilms.
- C) capsules.
- D) glycocalyx.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 23.6

45) Septicemia is an example of a

- A) systemic infection.
- B) transient infection.
- C) localized infection.
- D) autoimmune infection.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.2

46) The process by which white blood cells ingest and kill bacteria is called

- A) exocytosis.
- B) transcription.
- C) phagocytosis.
- D) translation.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.1

47) The condition that results when some bacteria are shed in the bloodstream and distributed to distant parts of the body but do not reproduce in the blood is called

- A) septicemia.
- B) bacteremia.
- C) uremia.
- D) erythemia.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.2

48) _____ are toxic proteins released from the pathogen during normal growth.

- A) Endotoxins
- B) Exotoxins
- C) Macrotoxins
- D) Microtoxins

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.6

49) Which of the following is a category of exotoxin?

- A) cytolytic toxin
- B) lipophilic toxin
- C) α -toxin
- D) β -toxin

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 25.6

50) _____ are microbial infections acquired by hospital patients with noninfectious diseases because they are compromised hosts.

- A) Healthcare-associated infections
- B) Immunocompromised infections
- C) Pathogenic infections
- D) Virulent infections

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.4

25.2 True/False Questions

1) Microbes initially colonize teeth by attaching to lipids attached to the enamel.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.2

2) Pathogenicity islands are clusters of genes for virulence factors that are located together on the bacterial chromosome.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.4

3) An opportunistic pathogen causes disease ONLY in the presence of normal host resistance.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.4

4) Virulence factors in *Salmonella* include antiphagocytic proteins.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.4

5) Virulence refers to the overall ability of an organism to cause disease.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.3

6) *Streptococcus mutans* produces dextran to aid in attachment to tooth surfaces.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.2

7) Pathogen virulence NEVER changes throughout a host's life.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.3

8) Cholera begins following a bite from a mosquito.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.6

9) Attenuated vaccines cause a weaker immune response than killed vaccines.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.3

10) *Streptococcus pneumonia* strains with capsules were called smooth because of their appearance on plates.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.3

11) *Vibrio cholerae* is a highly virulent pathogen compared with the influenza virus.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.3

12) Virulence genes may be located on mobile genetic elements.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.3

13) HIV infection causes immunosuppression because it causes the destruction of CD4 lymphocytes.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.4

14) R plasmids help prevent the spread of virulence factors.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.4

15) Invasion is the ability of a pathogen to enter into host cells or tissues, spread, and cause disease.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.2

16) The toxic lipopolysaccharides produced by most gram-negative bacteria are called exotoxins.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.8

17) Endotoxins are released in large amounts only when cells lyse.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.8

18) The *Limulus* amebocyte lysate assay is used to detect endotoxin in clinical samples such as serum or cerebrospinal fluid.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.8

19) Gram-positive bacteria produce endotoxins.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.8

20) Hormones that are produced under stress boost normal immune responses and play a role in stress-mediated disease resistance.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 23.12

21) Because malnutrition can make the immune system weaker, the number of *Vibrio cholera* cells necessary to produce cholera in an exposed individual is drastically reduced in individuals with substantial malnutrition.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 25.4

22) *Clostridium botulinum* are endospore-forming bacteria normally found in the soil.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.6

23) Hyaluronidase worsens infection by causing damage to red blood cells.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 25.5

24) Diphtheria toxin inactivates elongation factor 2 by catalyzing the attachment of adenosine diphosphate ribose from NAD⁺.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.6

25) *Clostridium botulinum* cannot produce toxins in improperly preserved foods.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 25.6

25.3 Essay Questions

1) Explain how diet plays a role in the development of dental caries.

Answer: Sugar in the human diet contributes to dental caries in two ways. First, sucrose (table sugar) is needed by *Streptococcus sobrinus* and *S.mutans* bacteria to produce capsules that they need to attach to teeth. Secondly, glucose is fermented and lactic acid is produced, which dissolves enamel and leads to the development of dental caries.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 25.2

2) It used to be thought that streptococci were the primary bacteria in dental plaque. What do we now know about the composition of dental plaque and how do we know it?

Answer: Dental plaque includes a diverse microbial community. It contains species of *Streptococcus*, *Corynebacterium*, *Porphyromonas*, *Lepotrichia*, *Neisseria*, *Fusobacterium*, and others. We know this through the use of phylogenetic probes. [Students may also describe more about FISH as explained in the introduction.]

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 25.2

3) When does an infection become a disease?

Answer: An infection becomes a disease when a microorganism that is established and growing in a host causes damage and injury that impairs host function.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.1

4) Design an experiment to determine whether a potential pathogen produces hemolysins.

Answer: Bacterial production of hemolysins is demonstrated in the laboratory by streaking the pathogen on blood agar plates containing 5% sheep's blood. If hemolysins are produced and released, red blood cell lysis will occur, releasing hemoglobin and creating a clear area called a zone of hemolysis around the growing colonies.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 25.7

5) Why does *Neisseria gonorrhea* adhere only to mucosal epithelial cells in the genitourinary tract, eye, rectum, and throat?

Answer: *N. gonorrhea* has a surface protein called Opa (opacity associated protein) that binds specifically to a host protein called CD66 that is found only on the surfaces of these cells.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.1

6) *Escherichia coli* is commonly found in the normal human microbiota, but some strains can cause severe illness. What is the difference between pathogenic and non-pathogenic *E. coli* in relation to the human host?

Answer: Most nonpathogenic *E. coli* strains are found in the cecum and colon and routinely pass through the body through the feces without causing disease. Pathogenic enterotoxigenic strains of *E. coli* contain genes encoding fimbrial colonization factor antigens, leading to bacterial adherence in the host small intestine. These bacteria produce enterotoxins, leading to diarrhea.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 25.1, 25.6

7) Explain how *Clostridium tetani* cells that rarely leave their initial site of infection can cause death in their hosts.

Answer: *C. tetani* produces tetanus toxins that move to distant parts of the body, initiating irreversible muscle contraction, and often death of the host. Although the cells do not actively travel around the host, the toxins it produces cause the death of the host.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.6

8) Describe the difference between the use of streptokinase and coagulase as a defense mechanism among pathogens.

Answer: Pathogens like *Streptococcus pyogenes* use streptokinase as a fibrinolytic enzyme that dissolves fibrin clots, which makes invasion of pathogens impossible by localizing their infection. On the other hand, coagulase is used by pathogens like *Staphylococcus aureus* to form an insoluble fibrin layer around them, thus protecting themselves from attacks from host cells.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.5

9) What are the different categories of exotoxins, and how do they function?

Answer: The three categories are AB toxins, cytolytic toxins, and superantigen toxins. The cytolytic toxins work by degrading cytoplasmic membrane integrity causing lysis. The superantigens work by stimulating large numbers of immune cells resulting in extensive inflammation and tissue damage. The AB toxins bind to host cell surface receptors and damage the cell when the A subunit is transferred across the targeted cytoplasmic membrane by the B subunit.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 25.6

10) Why is fever a common result of endotoxin exposure?

Answer: Endotoxins stimulate host cells to release cytokines that act as proteins that affect the temperature-controlling center of the brain, causing fever.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 25.8